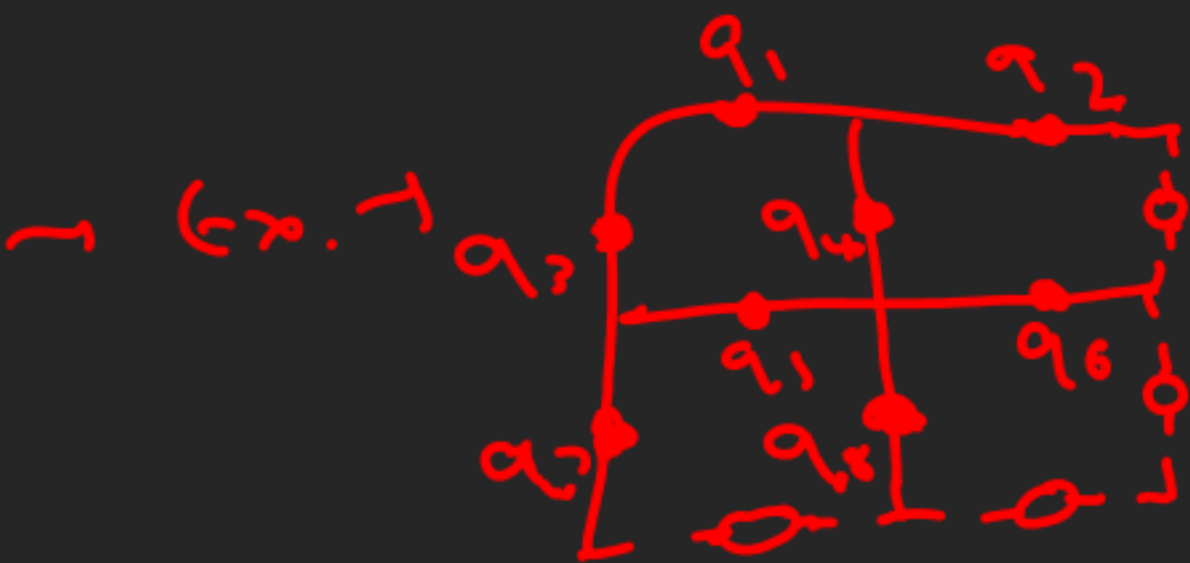
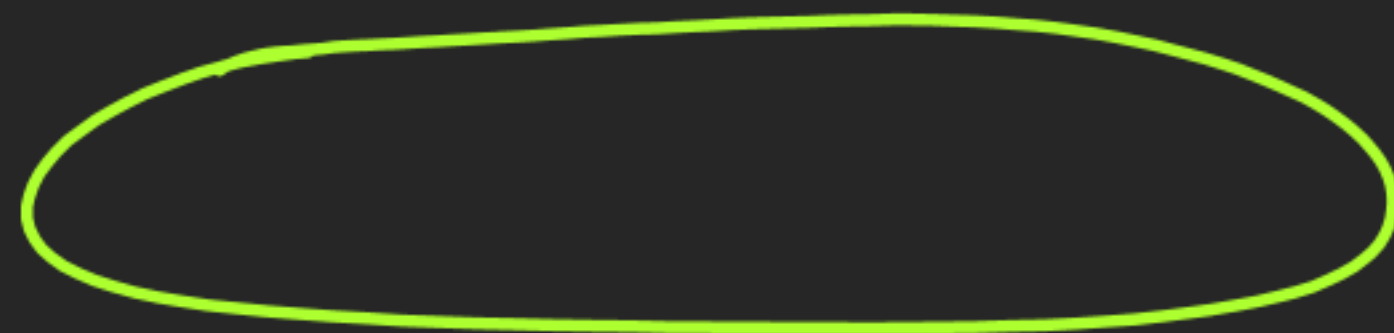




→ Forms X -sg



→ $Z \otimes I \otimes Z \otimes Z \otimes Z \otimes I \otimes I \otimes I$



$$\pi(Z\text{-sg}) = \pi(X\text{-sg}) = I$$

↓
All sg commute ea
other

$$(Z\text{-sg})_b = \prod_{a \neq b} (Z\text{-sg})_a$$

↪ If we remove 1 Z -sg it forms
minimal generating set

$$\begin{aligned} 2^{L^2-1} &\rightarrow Z\text{-sg} \\ 2^{L^2-1} &\rightarrow X\text{-sg} \\ 2^{L^2-2} &\rightarrow \text{sg} \end{aligned}$$

Total qubits $\rightarrow 2^{L^2} \rightarrow$ Encodes $2^{L^2 - (2^{L^2-2})}$
= 2 qubits

Detecting Errors

GitHub

(C_2, L^2, L, L)

Correcting Errors

GitHub